

A Measurably Self-Improving Multi-Agent Forecasting System: Five Feedback Loops, No Retraining

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Abstract

Decision-making under uncertainty is a defining capability of intelligent systems - and it is measurable: probability judgments can be scored against outcomes, and their error decomposed into bias and noise. What can be measured can be engineered.

We present a deployed multi-agent architecture for probabilistic forecasting that continuously improves through five closed feedback loops, each targeting a distinct source of prediction error, without any retraining. The system comprises seven specialized LLM forecasting agents, a meta-analyst (Cleon) that measures quality and routes corrections, and a knowledge-base agent (Cornelius) that turns surprising failures into insights served back before future predictions. Four loops are corrective and run downstream of a prediction: (1) noise-adaptive calibration feedback, routing to process-consistency guidance when noise dominates (93.5-99.7% of error in every agent); (2) surprise-driven insight extraction; (3) knowledge retrieval before prediction; and (4) hidden-state hypothesis injection. A fifth, generative loop runs upstream: synthesis abduces hidden-state theses from cross-domain anomalies, derives falsifiable consequences, and commissions them as real predictions - deciding what is worth forecasting rather than only scoring forecasts it was handed.

Deployed across 96,161 logged predictions and 73,552 resolved outcomes, the corrective loops show a measurable but split effect: three of five compared agents improved both Brier score and surprise rate (95% bootstrap CIs excluding zero), while two moved against their feedback - a limitation we report rather than hide. Against naive baselines on its own question set the fleet shows real but bounded skill (Brier 0.194 vs 0.250 always-0.5 and 0.217 base-rate; skill score +0.105 [+0.100, +0.110]); five of seven agents beat their own base rate, and published reference points (superforecasters 0.096, human crowds 0.149 to 0.19) come from different question sets.

The hidden-state store holds 56,347 scored hypotheses with a monotone win-rate gradient across lifecycle tiers (strong 75.0% > emerging 68.2% > fading 59.0%), validating the scoring. The generative loop is newly activated: its first three theses have commissioned all ten falsifiable consequences, with the earliest horizons resolving in September 2026. The system's aim is not certainty - under genuine uncertainty no forecaster delivers black-and-white answers - but demonstrable, measured skill: calibrated probabilities produced by a system that keeps score on itself, at fixed model capability.

1. Introduction

Decision-making under uncertainty is one of the defining capabilities of an intelligent system - and, contrary to its public image as an unmeasurable art, it is a measurable discipline. A probability judgment can be scored against the outcome it predicted (Brier 1950), the resulting error can be decomposed into systematic bias and noise (Murphy 1973; Kahneman et al. 2021), and the central lesson of the superforecasting tradition is that forecasters who keep score improve (Tetlock and Gardner 2015). Measurability is the premise of this paper: once forecasting error is measured and decomposed, each component becomes an engineering target - and an agentic system can be organized around feedback loops that target each component while the system runs.

Probabilistic forecasting is increasingly used as a benchmark and application domain for AI systems. Frontier LLMs now approach - but do not consistently match - human crowds and superforecasters on real forecasting questions (Halawi et al. 2024; Karger et al. 2024; Schoenegger, Tuminauskaite, et al. 2024), and bare models without scaffolding have performed no better than chance in live tournaments (Schoenegger and Park 2023). Most deployments nevertheless treat the forecaster as static: a model is prompted, generates a probability, and the result is logged. Improvement, if it happens at all, occurs through retraining (Turtel et al. 2025; Damani et al. 2025). This paper describes a different approach: a multi-agent system that continuously improves through structured feedback loops while agents are running, without any retraining.

The central architectural insight is that prediction error has multiple distinct sources - systematic bias, process inconsistency, wrong mental models, and ungrounded reasoning about the current state of the world - and each source requires a different correction mechanism. A single feedback signal (e.g., Brier score) conflates all of these. The distinction matters empirically: in our fleet, decomposition in the style of Kahneman et al. (2021) attributes 93.5-99.7% of error to noise (inconsistency), not bias - so the standard calibration toolkit (Guo et al. 2017; Geng et al. 2024) targets a minority component. Independent evidence points the same way: the same LLM asked the same forecasting question in different runs can answer below 20% and above 80% (FutureSearch et al. 2025). We decompose error and route each component to the appropriate intervention.

A second, more recent insight is that a scoring engine, however well-calibrated, has a structural blind spot: it can only improve forecasts it was asked to make. It has no theory-forming engine that decides what is worth forecasting in the first place. Our fifth loop closes exactly that gap by abduction from anomalies - generating hidden-state theses, deriving falsifiable consequences, and commissioning them as ordinary, honestly-resolved predictions.

The system has been running continuously since February 2026, processing a fleet of seven specialized LLM forecasting agents across domains from financial markets to AI semiconductors and biotech. The fleet's live record - predictions, resolutions, Brier scores, and hidden-state hypotheses - is published continuously at Brier HQ.¹ We present the architecture, describe each feedback loop, and report empirical evidence from 73,552 resolved predictions. We are deliberate about maturity: the four corrective loops are proven and produce measurable - if uneven - behavioral change,

¹<https://brierhq.ai> - the deployed system's public dashboard, updated as predictions are generated and resolved.

whereas the generative loop is newly activated and its first thesis-level outcomes do not resolve until September 2026. We report both honestly.

2. System Architecture

The system comprises three agent types connected by five feedback loops - four corrective (downstream of a prediction) and one generative (upstream, deciding what to predict). Figure 1 shows the architecture.

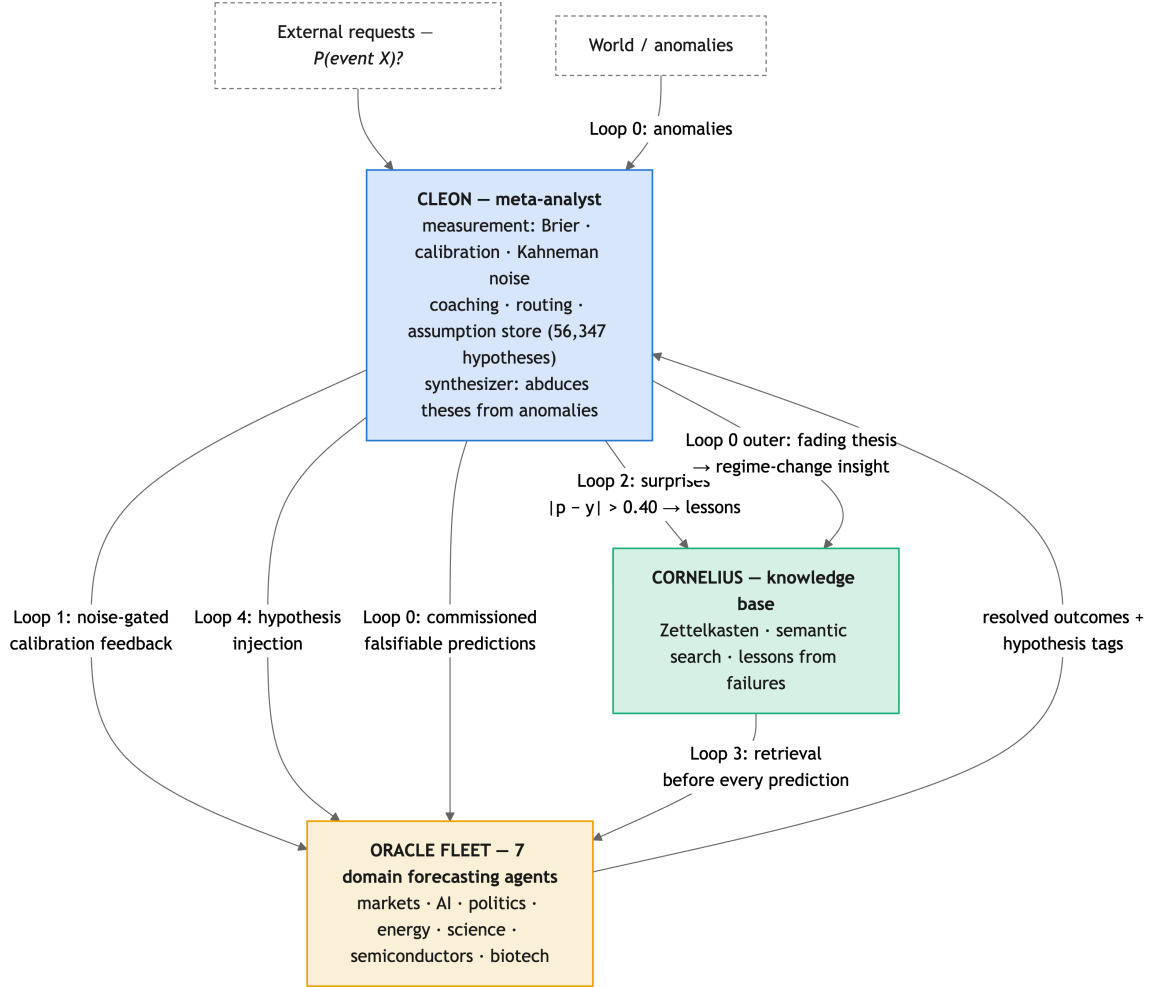


Figure 1: **Figure 1.** System architecture: the meta-analyst (Cleon), the seven-agent Oracle fleet, and the knowledge base (Cornelius), connected by five feedback loops - four corrective (1: noise-gated calibration feedback; 2: surprise-driven insight extraction; 3: retrieval before prediction; 4: hidden-state hypothesis injection) and one generative (0: abduction from anomalies to commissioned falsifiable predictions).

2.1 The Oracle Fleet

Seven LLM forecasting agents (Claude claude-opus-4-6), each specialized in a domain. Counts are from the 2026-07-10 analysis snapshot:

Agent	Domain	Logged	Resolved	Brier	Cal. err
oracle-2-markets	Financial Markets & Crypto	19,906	19,247	0.205	0.045
oracle-3-ai	Technology & AI	8,225	4,003	0.189	0.086
oracle-4-politics	Geopolitics & Politics	14,825	12,518	0.178	0.066
oracle-5-energy	Energy & Commodities	2,928	1,209	0.219	0.062
oracle-6-science	Space, Shipping & Climate	15,734	13,014	0.180	0.051
oracle-7-ai-semi	AI Semiconductors	15,373	10,139	0.214	0.062
oracle-8-biotech	Biotech, Pharma, FDA	19,170	13,422	0.191	0.101
Fleet		96,161	73,552		

All counts in this paper are pinned to the analyzer snapshot of 2026-07-10T10:02Z. Counts differ by up to ~150 rows (0.16%) across pipeline stages - the raw CSV corpus synced two hours later holds 96,308 logged / 73,559 resolved, and is the basis for the corpus-derived figures (Figures 3-5 and 7) - due to sync timing and malformed-row filtering. Prose and tables standardize on the analyzer snapshot.

The domain map reflects a June 2026 reorganization: oracle-5 was repurposed from geopolitics to Energy & Commodities (its prior history purged, so it is young - data begins June 2026, explaining its small n and higher Brier), and energy was carved out of oracle-6, which now covers Space, Shipping & Climate.

Each prediction records: statement, probability (0-1), domain, time horizon (short/medium/long), and chain-of-thought reasoning. All resolved outcomes are binary (0 or 1). Predictions are written through a validated CSV pipeline that enforces field quoting, duplicate prevention, status/outcome consistency, and - since June 2026 - a resolution source-trust tier, preventing the column corruption and unsourced-resolution drift that afflicted earlier versions of the system.

All agents are deployed on Trinity, an open-source self-hosted runtime for autonomous CLI agents.² Each agent runs two autonomous schedules: `/resolve-backlog` (every 2h, processes up to 30 overdue predictions against tiered sources) and `/generate` (every 4h, generates 10-25 new predictions after querying Cornelius and loading calibration notes). This separation keeps resolved-prediction context from interfering with generation quality. Figure 2 shows the fleet live on the runtime's operations dashboard - the staggered schedule slots and the Loop 3 query traffic into Cornelius are directly visible.

²<https://github.com/Abilityai/trinity> (Apache 2.0).

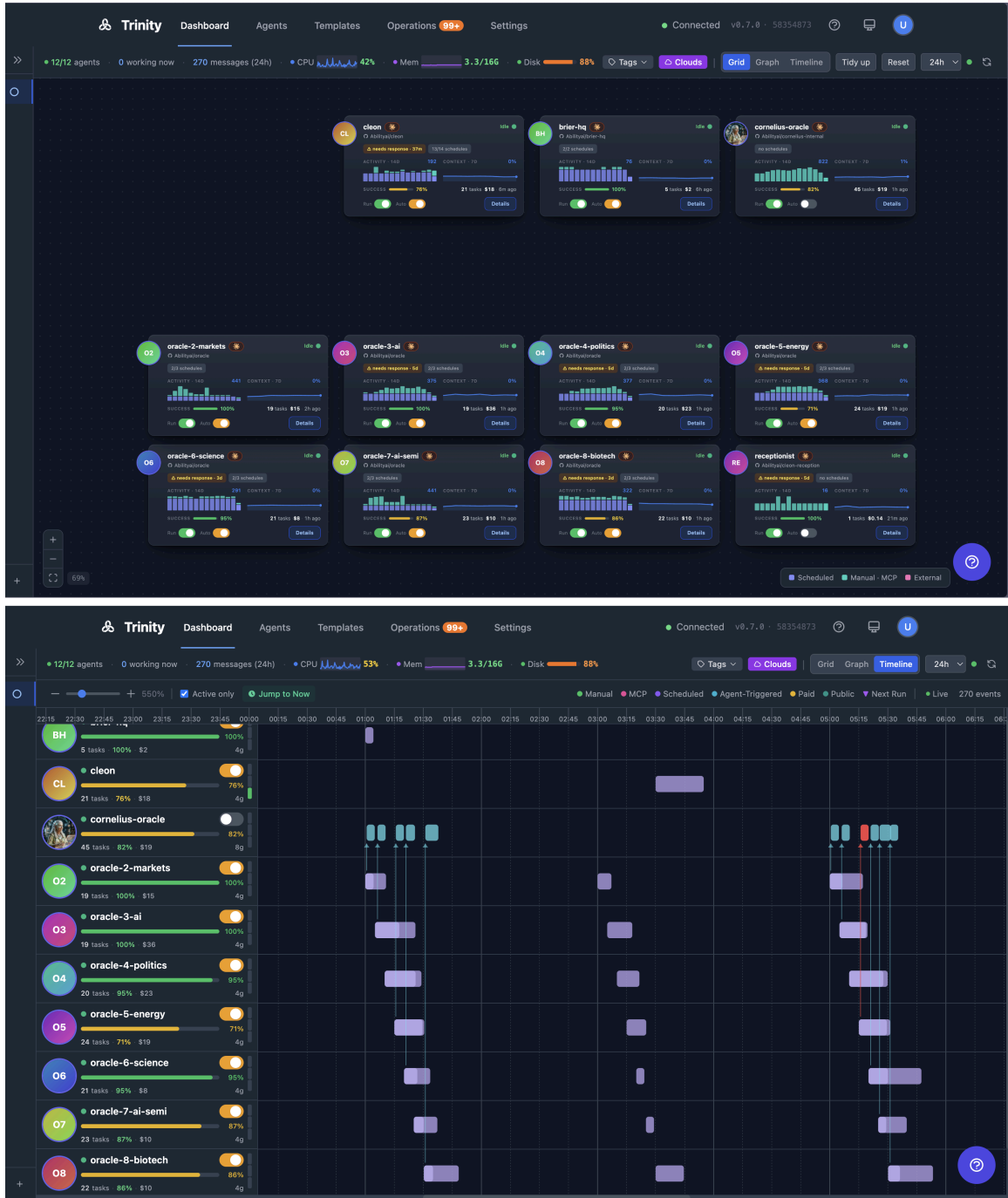


Figure 2: **Figure 2.** The deployed system on the Trinity runtime (operations dashboard, screenshots dated 2026-07-12). Top: the agent grid - the seven Oracle forecasting agents alongside the meta-analyst (clean), the knowledge-base agent (cornelius), and auxiliary agents including the public Brier HQ publisher - with 14-day activity, task counts, success rates, and schedule status per agent. Bottom: a 24-hour operations timeline; purple blocks are the staggered scheduled runs described in Section 2.1 (each Oracle's /resolve-backlog and /generate slots), and the teal arrows are agent-triggered calls from the Oracles' runs into cornelius - the retrieval-before-prediction queries of Loop 3 (Section 6) occurring live.

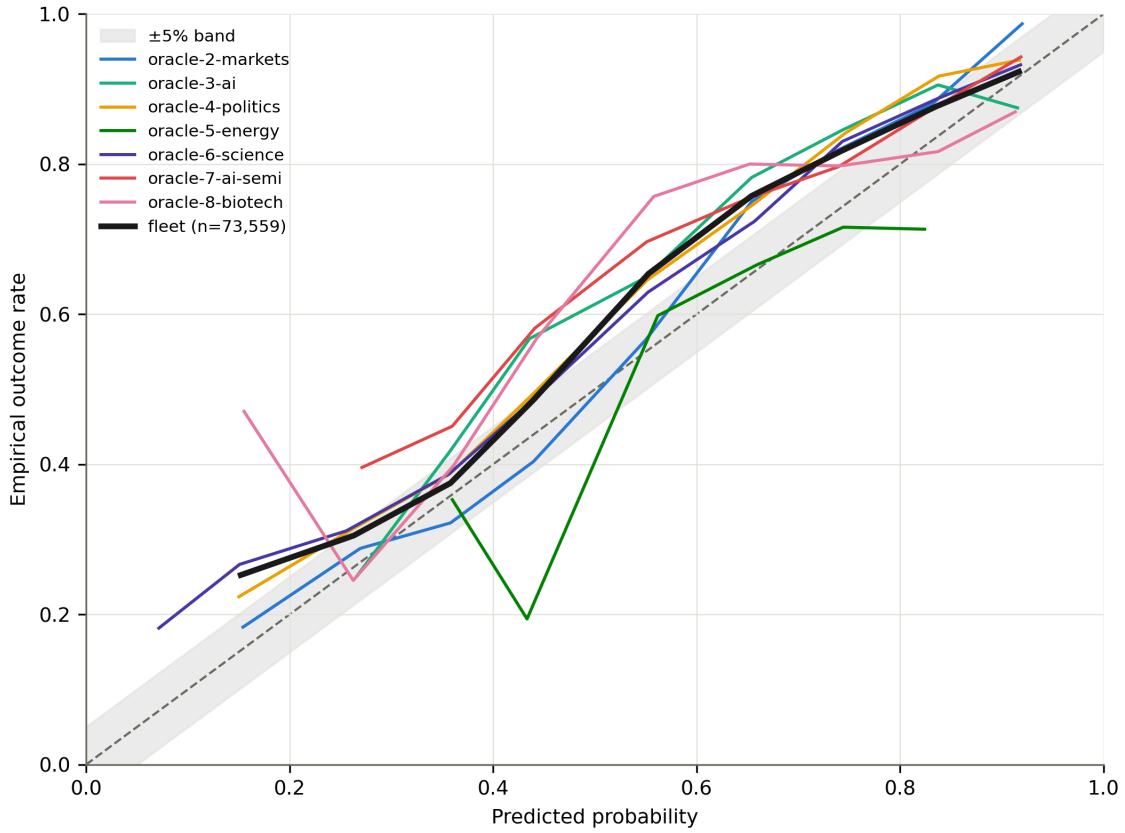


Figure 3: **Figure 3.** Fleet calibration curve computed from the raw prediction corpus (synced 2026-07-10; 73,559 resolved rows; bins with $n \geq 30$ per agent, $n \geq 100$ fleet; analysis/generate_figures.py). The black line is the fleet aggregate; colored lines are individual Oracles. The fleet tracks the diagonal closely across most of the probability range; the young oracle-5 (green) is the visible outlier at mid-range.

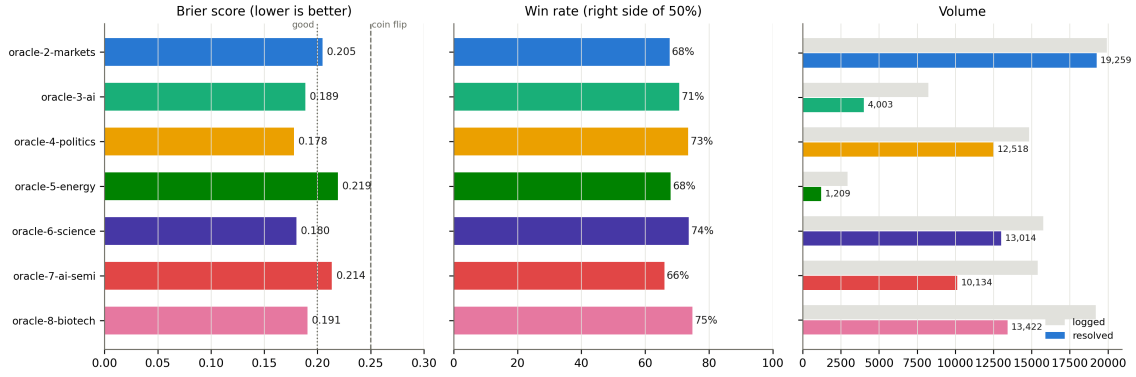


Figure 4: **Figure 4.** Per-agent Brier score (left; lower is better, with “coin flip” and “good” thresholds marked), win rate (center; right side of 50%), and volume (right, logged vs resolved), computed from the raw corpus (`analysis/generate_figures.py`). oracle-4-politics has the best Brier (0.178); the young oracle-5-energy the weakest (0.219); oracle-2-markets carries the largest volume.

2.2 Cleon - Analyst, Coach, Router, and Synthesizer

Cleon is the meta-analyst agent. Its responsibilities:

Measurement - Computes Brier scores, calibration error, surprise rates, and a noise decomposition in the style of Kahneman et al. (2021) (level bias, pattern noise, occasion noise, conditional bias) per agent across (domain x time-horizon) cells. Analysis runs daily from synced CSVs and is schema-validated on write and read, so a shape change fails loud at the seam rather than silently degrading downstream consumers.

Coaching (Loop 1) - Generates and delivers calibration feedback to each Oracle as advisory text (`calibration-notes.md`), read at prediction time. Uses a noise-dominance gate (Section 4) to route between process-consistency guidance and probability adjustment.

Routing - Receives external prediction requests, identifies the domain, injects relevant hidden-state hypotheses, routes to the Oracle with the best domain track record, and returns the prediction with provenance.

Hidden-State Tracking (Loop 4) - Maintains the assumption store: a FAISS-indexed collection of world-state hypotheses scored by prediction outcomes (Section 7).

Synthesis (Loop 0) - Abduces cross-domain hidden-state theses from world anomalies, derives falsifiable consequences, and commissions them as real predictions (Section 3). This is the one component that decides *what* to forecast rather than scoring forecasts it was handed.

2.3 Cornelius - The Knowledge Base

Cornelius is a Zettelkasten-based knowledge-base agent combining a graph structure with semantic vector search, in the spirit of retrieval-augmented generation (Lewis et al. 2020) and LLM-knowledge-graph integration (S. Pan et al. 2024). An open-source template of its architecture

is publicly available.³ It plays two roles in the feedback architecture:

As an insight sink (Loop 2): When a prediction fails with surprise score > 0.40 (i.e., $|\text{probability} - \text{outcome}| > 0.40$), Cleon sends the full context - prediction, reasoning, outcome - to Cornelius. Cornelius extracts a durable insight: not “Oracle-X was wrong about Y” but a transferable lesson about *how the world works* that the agent’s reasoning failed to account for. For example:

“OPEC+ can execute emergency output changes faster than the assumed 3-month lead time. The model’s supply-side response lag is systematically wrong.”

These insights are stored as permanent notes tagged by domain and failure type, forming a growing corpus of lessons from the fleet’s collective mistakes.

As a knowledge source (Loop 3): Before each Oracle generates any prediction, it queries Cornelius via semantic search: “What do we know about [topic/domain]?” Cornelius returns the most relevant prior insights. The Oracle incorporates these into its reasoning before generating a probability.

This creates a learning dynamic: one Oracle’s failure, once processed by Cornelius, becomes every Oracle’s lesson on their next relevant prediction. The fleet learns collectively without sharing state directly.

2.4 Baselines and Reference Points

Raw Brier scores (Brier 1950) are difficult to interpret without a baseline: their attainable range depends on the outcome base rate of the question set, and our corpus is imbalanced - 68.3% of resolved predictions resolve *true* (per-agent base rates 0.608-0.761), reflecting the fleet’s self-generated question mix. We therefore anchor the fleet’s accuracy two ways: against naive statistical baselines computed on *exactly the same questions*, and against published human and LLM reference points from the literature, which come from *different question sets* and are context rather than comparison.

Internal baselines (same questions). We compare each agent against (a) a uniform forecaster that always predicts 0.5 (Brier 0.250 by construction) and (b) an in-sample base-rate forecaster that always predicts the agent’s own empirical outcome frequency - the climatological baseline underlying the standard skill-score and calibration-resolution decomposition (Murphy 1973). The base-rate baseline is deliberately strong: it is granted oracle knowledge of the final base rate, which no live forecaster has. We report the Brier Skill Score, $BSS = 1 - \text{Brier}/\text{Brier}_{\text{baseline}}$ (positive = skill beyond the baseline), with 95% bootstrap CIs (percentile, 2,000 resamples), computed from the raw corpus (73,559 resolved rows, synced 2026-07-10; analysis/baseline_comparison.py):

³<https://github.com/Abilityai/cornelius> - the second-brain agent template; the deployed instance adds the fleet-specific insight corpus.

Agent	n	Base rate	Brier	Base-rate Brier	BSS vs base rate (95% CI)
oracle-2-markets	19,259	0.608	0.205	0.238	+0.141 [+0.133, +0.149]
oracle-3-ai	4,003	0.699	0.189	0.210	+0.104 [+0.076, +0.128]
oracle-4-politics	12,518	0.698	0.178	0.211	+0.156 [+0.140, +0.170]
oracle-5-energy	1,209	0.631	0.219	0.233	+0.058 [+0.026, +0.089]
oracle-6-science	13,014	0.691	0.180	0.214	+0.157 [+0.143, +0.171]
oracle-7-ai-semi	10,134	0.693	0.214	0.213	-0.003 [-0.018, +0.012]
oracle-8-biotech	13,422	0.761	0.191	0.182	-0.047 [-0.063, -0.033]
Fleet	73,559	0.683	0.194	0.217	+0.105 [+0.100, +0.110]

Every agent clears the uniform baseline decisively (fleet BSS vs uniform +0.225). The base-rate baseline is more discriminating, and the result is honest rather than flattering: five of seven agents show real skill beyond their own base rate (BSS +0.058 to +0.157, all CIs excluding zero), the fleet as a whole scores +0.105 [+0.100, +0.110], but oracle-7-ai-semi is statistically indistinguishable from its base rate (-0.003 [-0.018, +0.012]) and oracle-8-biotech falls significantly *below* it (-0.047 [-0.063, -0.033]). oracle-8’s case is instructive: its raw Brier (0.191) looks mid-fleet, but 76.1% of its questions resolve true, so simply predicting 0.761 everywhere would have scored better - raw Brier flatters agents on lopsided question mixes, which is exactly why baseline anchoring matters. The two results connect to the Loop 1 evidence (Section 4.3) in different ways: oracle-7, which moved against its calibration feedback, is also skill-less against its base rate - the same agent surfaces as the problem case on both yardsticks - while oracle-8, the loop’s clearest success (confidence +7.7pp, surprise -11.9pp, Brier improving), is demonstrably moving *toward* a baseline it has not yet crossed.

External reference points (different questions). Published Brier scores locate the fleet’s operating range in the literature (Figure 5, right panel): superforecasters score 0.096, the general public 0.121, and the best LLM 0.122 on ForecastBench’s contamination-free question set (Karger et al. 2024); the human crowd scores 0.149 and a retrieval-augmented LLM system 0.179 on five forecasting platforms, with GPT-4 without retrieval at 0.208 (Halawi et al. 2024); a 12-LLM ensemble scores 0.20 against a human crowd’s 0.19 in a live Metaculus tournament (Schoenegger, Tuminauskaite, et al. 2024); and a 14B model RL-trained on resolved outcomes reaches 0.190 (Turtel et al. 2025). Our fleet’s 0.194 sits in the band occupied by current LLM forecasting systems and above (worse than) human crowds and superforecasters. We emphasize that these numbers are **not directly comparable** to ours: Brier scores are only comparable on a common question set, and question difficulty, base rates, and resolution procedures all differ. Evaluating this fleet

on a standing external benchmark such as ForecastBench is future work (Section 10); the internal skill scores above are the apples-to-apples evaluation this paper stands on.

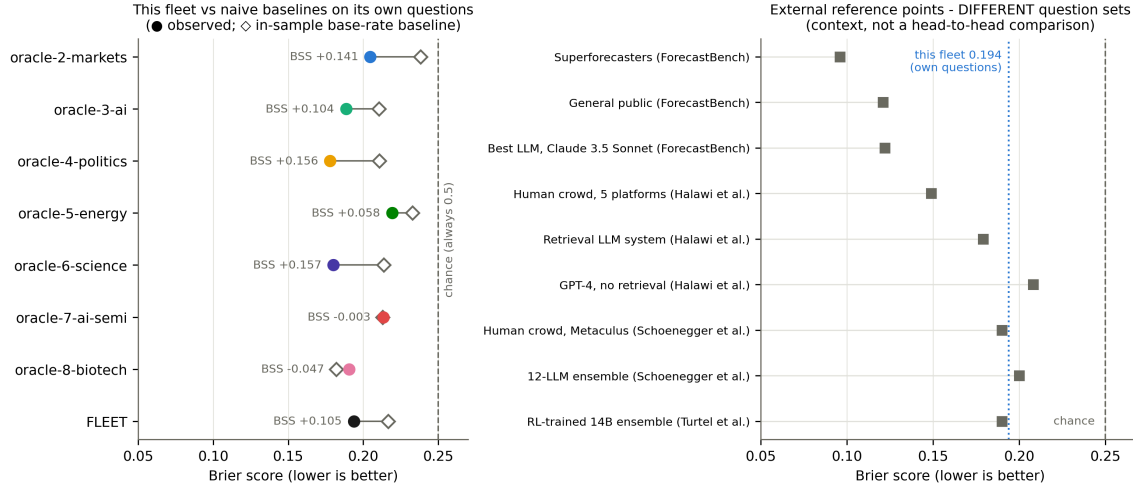


Figure 5: **Figure 5.** Baselines and reference points. Left: per-agent and fleet Brier scores (filled circles) vs the in-sample base-rate baseline (open diamonds) on the fleet’s own questions, with Brier Skill Scores; the dashed line is the always-0.5 chance baseline (0.250). Five of seven agents beat their base rate; oracle-7 ties it; oracle-8 falls below it. Right: published Brier scores from the literature - different question sets, shown as context, not head-to-head (superforecasters, public, and best LLM from ForecastBench (Karger et al. 2024); crowd, system, and GPT-4 from Halawi et al. (2024); Metaculus crowd and LLM ensemble from Schoenegger, Tuminauskaite, et al. (2024); RL-trained 14B from Turtel et al. (2025)). Computed from the raw corpus by analysis/baseline_comparison.py.

3. Loop 0: Synthesis - Deciding What to Forecast

3.1 Why a Generative Loop

Loops 1-4 are *corrective*: they measure and repair forecasts the fleet was handed. Loop 0 runs the other way - **world** -> **anomaly** -> **hidden cause** -> **consequences** -> **commissioned forecasts** - and decides *what is worth forecasting*, which nothing else in the system does.

The motivation is a structural blind spot. Every hypothesis in the assumption store (Loop 4) is *extracted from* forecasts the Oracles were already asked to make. The store is therefore a **mirror of what we chose to ask, not a searchlight on the world**. A regime shift nobody is already forecasting - for example, “central banks are quietly de-dollarizing into physical gold and building a parallel settlement system” - never enters the store, because no prediction pointed at it. A blind test on 2026-07-01 confirmed this: querying the store for the four pillars of a circulating China/gold/de-dollarization thesis returned only unrelated noise. The missing operation is **abduction** - inferring a non-obvious present state that explains a cluster of surface events, then deriving what *else* would be true if it held. Abductive event-explanation has begun to appear in the forecasting

literature (Shi et al. 2023), and LLMs have shown they can propose genuinely novel hypotheses (Qi et al. 2023); Loop 0 wires that capability onto an honest resolution back-end.

3.2 Mechanism

The loop has five stages; the inner loop is fully built and closed.

1. **Anomaly detection** (/anomaly-scan) - scans Cleon's local substrate (the synced prediction corpus, the surprise log, and the assumption store) for salient, cross-domain events the current model does not explain, and emits a ranked anomaly-cluster artifact. The trigger - surprise relative to the system's own beliefs - echoes surprise-driven discovery search (Agarwal et al. 2025).
2. **Abductive generation** (/synthesize) - for each anomaly cluster, generates candidate hidden-state **theses** that would explain all of it at once, each paired with its competing "official / null" explanation. Explicitly cross-domain, spanning markets + geopolitics + commodities, which no single vertical Oracle can see.
3. **Consequence derivation** (/synthesize) - for each thesis, derives at least three falsifiable observable implications: "*if this thesis is true, we would also see X, Y, Z within horizon H.*" Each carries a null-differential and resolution criterion, gated by `validate_theses.py` so a null-less or unfalsifiable thesis fails loud.
4. **Commissioned predictions** (/commission-theses) - turns each consequence into a concrete prediction routed to the best Oracle via the existing /predict path, tagged with `thesis_id/ consequence_id` so the resolved outcome flows back into the thesis store. A thesis thus becomes a *bundle of falsifiable forecasts*, reusing the entire prediction -> resolution -> scoring machinery.
5. **Thesis tracking** (/thesis-store) - each thesis is a first-class object scored by the win/loss of its commissioned predictions, at a higher altitude than a single hypothesis (cross-domain, long-horizon). Reset-and-replay scoring with 30-day recency weighting drives an emerging -> supported -> fading -> dormant lifecycle. A fading thesis is the assumption store's regime-change signal *one level up*, surfaced as an alert and never deleted.

The outer loop - feeding insights back. A *supported* thesis promotes its hidden state into the assumption store as an Oracle-injected hypothesis (source: `thesis:<id>`), so a validated insight actually improves the fleet's *next* forecast - and it **retracts** if the thesis later fades, so no stale belief lingers. A *fading* thesis is delivered to Cornelius as a regime-change insight, propagating to every Oracle via Loop 3.

3.3 Status: Wired and Testing

As of the 2026-07-10 snapshot, Loop 0 has crossed from built-but-idle to **wired and testing**, but has earned no thesis-level calibration yet:

Metric	Value
Theses in store	3 (all emerging)
Consequences	10 total, 10 commissioned , 0 resolved
Consequences per thesis	3.33 (meets the ≥ 3 falsifiable-by-design invariant)
Commission-wire integrity	verified - all 10 have matching source: loop0-commission predictions
Earliest consequence horizon	2026-09-19

The three theses are iran-hormuz-toll-regime (4 consequences), us-economic-security-campaign (3), and tighter-for-longer-us-energy-rates (3). Stage 4’s first commissioning run converted every consequence into a real, tagged fleet prediction; the commission wire verified intact. **No thesis win rates are reported because none have been earned** - the earliest horizons resolve in September 2026, at which point the daily re-score converts the layer from “wired” to “scored.”

What this buys us that a narrative analyst cannot have: a human commentator produces bold, non-consensus theses but never has to settle them - no calibration, no scorecard. Loop 0 wires an abductive front-end onto our honest-resolution back-end: **generate theses as bold as a human analyst’s, then kill the wrong ones with data in weeks.**

4. Loop 1: Noise-Adaptive Calibration Feedback

4.1 Error Decomposition

For each agent, we apply the noise framework of Kahneman et al. (2021): group predictions into (domain x time-horizon) cells, then compute:

- **Pattern noise:** within-cell variance of Brier scores - inconsistency on structurally similar questions
- **Occasion noise:** temporal drift within cells - quality changing over time for the same question type
- **Conditional bias:** variance of cell-level biases - whether error direction varies by (domain, horizon)
- **Level bias:** mean(predictions) - mean(outcomes) - systematic over/underconfidence

This mirrors the classic decomposition of judgment error into bias and variance components (Gigerenzer and Brighton 2009; Yates et al. 1989). The result is consistent across all seven agents: noise accounts for **93.5-99.7%** of prediction error; level bias is 0.3-6.5%. The dominant noise sources are occasion noise (temporal quality drift) and conditional bias (error direction varies by question type); pattern noise is uniformly low.

Agent	Bias%	Noise%	Pattern	Occasion	Cond. bias (var)
oracle-2-markets	0.5%	99.5%	0.018	0.021	0.0067
oracle-3-ai	4.6%	95.4%	0.023	0.029	0.0125
oracle-4-politics	3.3%	96.7%	0.027	0.029	0.0049
oracle-5-energy	0.3%	99.7%	0.034	0.047	0.0216
oracle-6-science	2.3%	97.7%	0.031	0.044	0.0080
oracle-7-ai-semi	5.9%	94.1%	0.017	0.052	0.0185
oracle-8-biotech	6.5%	93.5%	0.025	0.069	0.0209

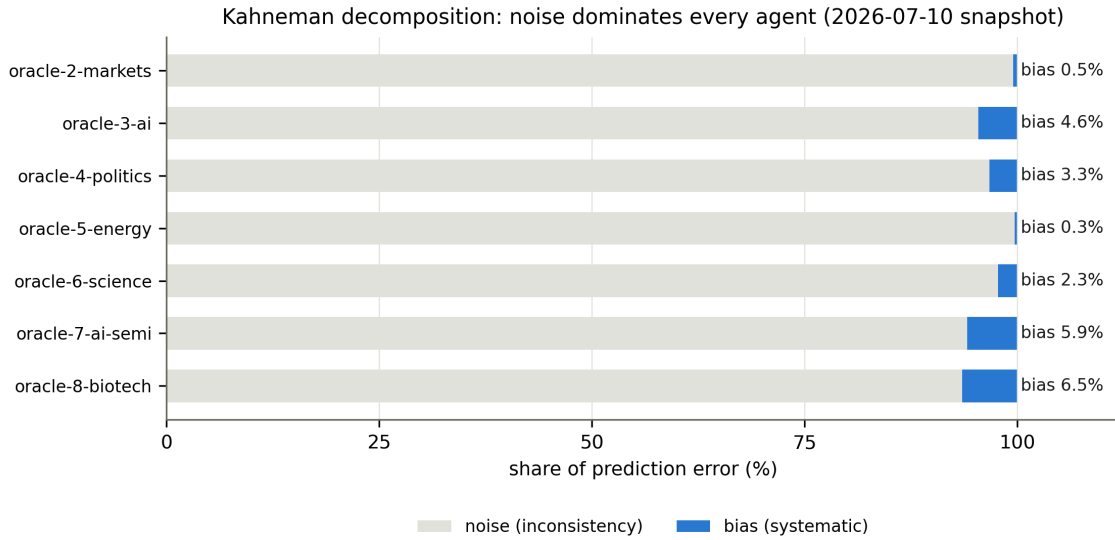


Figure 6: **Figure 6.** The decomposition per agent (2026-07-10 analysis snapshot, latest-analysis.json): noise accounts for 93.5-99.7% of prediction error in every agent; the systematic-bias share (blue) never exceeds 6.5%.

This is not unique to our fleet (Figure 6): run-to-run inconsistency has been independently documented as the dominant variance source in LLM forecasting, with the same model answering the same question below 20% in one run and above 80% in another, and roughly 60% of variance attributable to final judgment rather than evidence gathering (FutureSearch et al. 2025).

4.2 The Noise-Dominance Gate

This finding shapes the feedback design. Standard calibration techniques - bucket recalibration, temperature scaling (Guo et al. 2017), confidence elicitation tuning (Tian et al. 2023; Lin et al. 2022) - target the bias component. Applying them to a noise-dominated system addresses the wrong error source.

We implement a noise-dominance gate:

```

if noise_pct > 0.80:          # Current state for all agents
    -> process consistency guidance
        structured reasoning templates for high-noise cells
        temporal drift alerts when occasion noise > 0.05
        per-category conditional bias adjustments (not global nudges)
else:
    -> bucket probability adjustments
        systematic bias corrections

```

Per-category conditional-bias adjustment is notable: because bias direction varies by (domain, horizon), a global “you’re overconfident” message helps some cells while actively hurting others. The gate routes to targeted per-cell guidance instead.

Feedback is rate-limited (7-day cooldown) and diff-checked (suppress resends of identical content) to prevent the feedback loop from itself becoming a noise source. It is also gated on fleet health: no calibration is pushed onto a fleet whose data-quality check is red.

4.3 Evidence: A Split Fleet

Since calibration feedback began (first deliveries 2026-03-24), the fleet shows a *split* - the honest headline is that the loop works where its advice is absorbed, and two agents move against it. On the fleet before/after comparison (Figure 7), three of five compared agents improved both Brier score and surprise rate, with 95% bootstrap confidence intervals excluding zero on both metrics; a fourth (oracle-3) moved in the same direction without reaching significance:

Agent	Feedback		Δ surprise (95% CI)	Δ Brier (95% CI)	Mean prob. shift
oracle-2-mar-kets	raise confidence	confi-	+5.2pp [+3.4, +7.0]	+0.012 [+0.007, +0.017]	-2.5pp
oracle-3-ai	(near-calibrated)		-9.2pp [-17.9, +0.0]	-0.008 [-0.030, +0.013]	+0.7pp
oracle-4-politics	raise confidence	confi-	-12.8pp [-15.2, -10.3]	-0.028 [-0.035, -0.021]	+5.6pp
oracle-6-science	consistency		-14.7pp [-17.6, -11.4]	-0.019 [-0.029, -0.010]	+1.0pp
oracle-7-ai-semi	raise confidence	confi-	+7.1pp [+3.9, +10.4]	+0.009 [-0.001, +0.018]	+1.4pp
oracle-8-biotech	raise ~11pp		-11.9pp [-16.7, -6.7]	-0.020 [-0.033, -0.008]	+7.7pp

Fleet-wide surprise dropped 48.7% -> 44.4% (-4.3pp), and every agent holds calibration error in the 0.045-0.101 band. (Deltas are after minus before at each agent’s first feedback delivery: 2026-03-24 for the five agents whose history predates feedback; oracle-8 launched 2026-04-03 and splits at its first feedback cycle, 2026-04-05. Bootstrap CIs: percentile, 2,000 resamples. Internal

system-effectiveness reports use a uniform 2026-04-05 split and show slightly different deltas, e.g. oracle-4 -13.3pp; the qualitative split-fleet pattern is identical under both splits. oracle-5-energy is excluded because its history begins June 2026.)

The cleanest causal chain is **oracle-8-biotech**. Told repeatedly it was ~11pp underconfident, it raised mean confidence 57.3% -> 65.0% (+7.7pp), with *every* biotech sub-domain moving up (biotech +20.2pp, clinical-trials +8.8pp, FDA +7.5pp, pharma +3.3pp), cut its surprise rate 53.4% -> 41.5% (-11.9pp [-16.7, -6.7]), and improved its Brier score (-0.020 [-0.033, -0.008]) while making that shift. Feedback direction matched on every axis - the archetype the loop is designed to produce.

The counter-signal is itself informative. **oracle-2-markets** and **oracle-7-ai-semi**, both told to raise confidence, did the opposite (oracle-2 lowered it across crypto, markets, and economics) and saw surprise *rise*. The loop delivers advisory text, and absorption is not guaranteed. That two of seven agents move against their feedback is a limitation we report rather than hide; it points to per-agent intervention design - deciding when a global advisory should become a per-domain correction, or when an agent needs a different mechanism entirely - as the concrete next step.

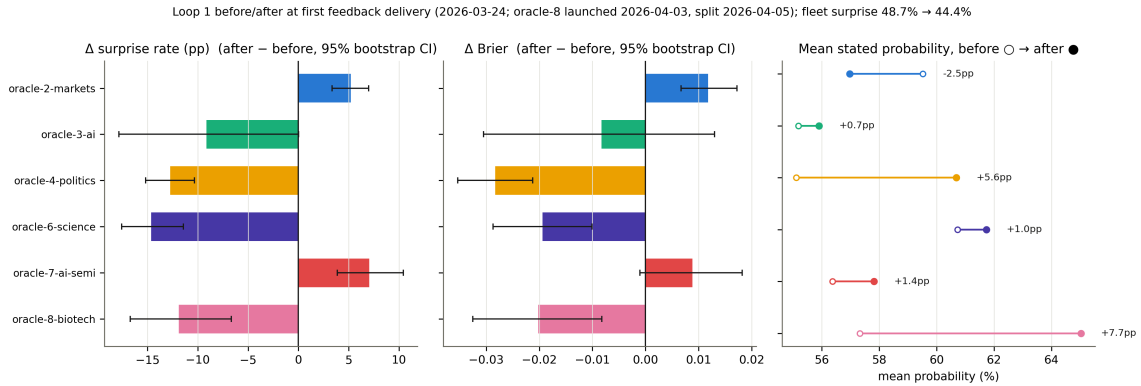


Figure 7: **Figure 7.** Loop 1 before/after evidence, computed from the raw corpus at each agent’s first feedback delivery (2026-03-24; oracle-8: 2026-04-05) by `analysis/generate_figures.py`. Left: change in surprise rate with 95% bootstrap CIs (2,000 resamples). Center: change in Brier score with CIs. Right: mean stated probability before (open) vs after (filled). oracle-4, oracle-6, and oracle-8 improve on both metrics with CIs excluding zero; oracle-2 and oracle-7 regress against their feedback.

5. Loop 2: Surprise-Driven Insight Extraction

After resolved predictions are synced, Cleon scans for surprises: $|\text{probability} - \text{outcome}| > 0.40$. For each surprise, the full context (prediction statement, reasoning, outcome, resolution notes) is sent to Cornelius.

Cornelius applies a structured extraction prompt that focuses on *what assumption failed*, not *which agent was wrong*:

“Oracle predicted [statement] at [confidence]%. The outcome was [opposite]. The reasoning was: [reasoning]. What did the model get wrong about how the world works? What assumption failed?”

The stored insight is domain-tagged (`#prediction-failure #[domain]`) and becomes permanently retrievable, growing the corpus continuously as the fleet resolves new predictions. This failure-to-lesson distillation is the deployed analogue of reflection-based agent learning (Shinn et al. 2023; Zhao et al. 2024; Gupta et al. 2024), with one critical difference: the trigger is a *resolved real-world outcome*, not a model’s own judgment of its output.

Delivery is asynchronous and reconciled: each batch is dispatched, recorded in-flight with its execution ID, and confirmed on the next run - with automatic requeue on failure. Detection outpaces delivery by a wide margin (Figure 8): as of 2026-07-09, the pipeline had processed 27,818 surprise-scan entries cumulatively, of which 1,967 insights were delivered to Cornelius; 390 remained pending retry, a queue that is rising and concentrated in the highest-volume agent (343 of 390 in oracle-2, driven by delivery timeouts). The health monitor tracks this as a rising-queue trend rather than letting it fail silently.

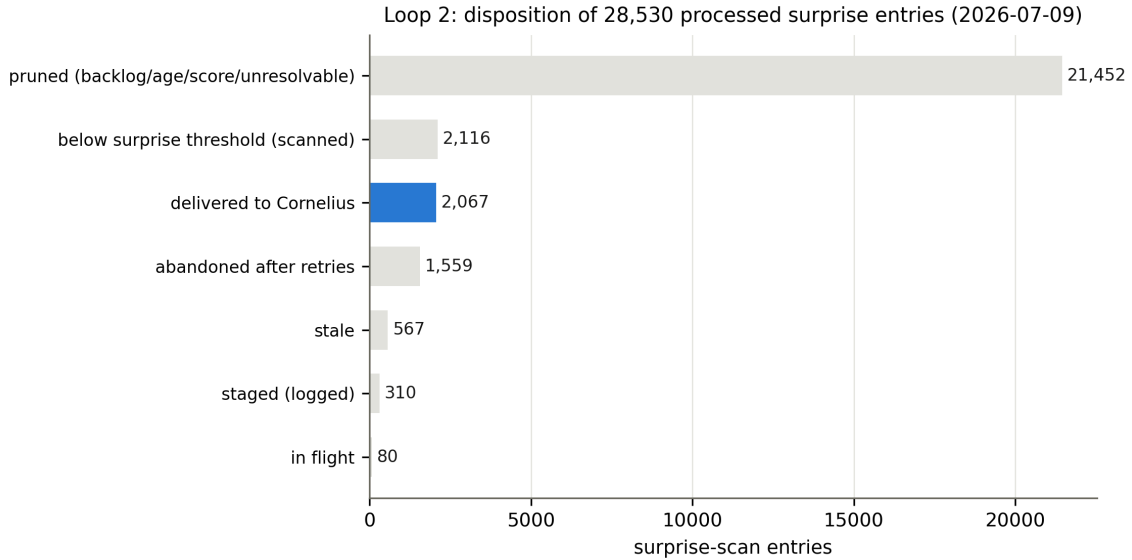


Figure 8: **Figure 8.** Loop 2 delivery funnel: disposition of every entry in the surprise log (2026-07-09), counted directly from the log by `analysis/generate_figures.py`. Pruning (backlog, age, score, unresolvable) dominates; delivered insights (blue) are a small slice of detected surprises - the measured effect of Loops 2-3 is therefore a lower bound on the design’s potential.

6. Loop 3: Knowledge Retrieval Before Prediction

Before generating any probability, each Oracle issues a semantic query to Cornelius:

“What insights do we have about [prediction topic / domain]?”

Cornelius returns the top-k relevant notes via 3-layer semantic search (keyword, embedding, graph traversal). The Oracle reads these before reasoning - incorporating lessons from prior failures of itself and all other fleet members. Retrieval before generation is a well-established reliability lever (Lewis et al. 2020; Gao et al. 2023), and interleaving it with reasoning measurably reduces factual error (Trivedi et al. 2023; Jiang et al. 2023); the distinctive property here is that the retrieval corpus is *self-generated from the system’s own failures* rather than an external document collection, and shared experience demonstrably transfers across agents (Qian et al. 2024).

This is the mechanism by which one Oracle’s mistake becomes every Oracle’s safeguard. An oracle-5 failure about OPEC+ supply-response speed, once extracted and stored by Cornelius, prevents oracle-6 from making the same misjudgment when predicting LNG shipping routes. The fleet learns collectively without any direct agent-to-agent knowledge transfer.

7. Loop 4: Hidden-State Hypothesis Injection

7.1 The Core Insight

Every probabilistic prediction is implicitly a bet on the current (non-observable) state of reality. “Oil will exceed \$100/barrel in Q3 at 75%” is not just a price model - it implicitly assumes that specific hidden conditions obtain right now. When a prediction fails, it may be because a hidden-state assumption was wrong - not because the probability model was miscalibrated. Standard calibration feedback cannot distinguish these failure modes.

Maintaining an explicit, probabilistic representation of unobservable state has a long lineage in dialogue systems, where belief trackers score distributions over hidden, non-stationary user goals (Henderson et al. 2014; Williams et al. 2016; Mrkšić et al. 2017). The assumption store is a forecasting analogue at world scale: hypotheses about current reality, scored by predictive outcomes rather than dialogue consistency.

7.2 Extraction and Scoring

Extraction: After each resolution, an LLM extracts hidden-state hypotheses from the prediction’s reasoning. Each hypothesis receives a *counterfactual shift estimate*: how many pp the prediction probability would change if the hypothesis were false. Hypotheses with shift < 5pp are rejected as truisms (e.g., “no major exchange is currently insolvent” - almost always true, adds no predictive information).

Deduplication: New hypotheses are compared against the store via FAISS cosine similarity (all-MiniLM-L6-v2, 384-dim). Hypotheses within 0.80 cosine distance of an existing entry are merged; those exceeding the threshold are added. The store rebuilds incrementally - a hypothesis ID is a content hash of its text, so its vector is immutable and cached, keeping each rebuild O(new work) rather than O(total corpus).

Scoring: Each prediction tags which hypothesis IDs it relied on. When it resolves, win/loss counters update. Recency weighting (30-day exponential half-life) enables regime-change detection:

```
weight(days_ago) = 0.5 ** (days_ago / 30)
recent_confidence = weighted_wins / (weighted_wins + weighted_losses)
```

Status lifecycle:

- emerging: < 5 uses - too few to judge
- strong: recent_confidence ≥ 0.55 and ≥ 5 uses - validated
- fading: recent_confidence < 0.50 and ≥ 10 uses - **regime-change signal**
- dormant: not selected in 60+ days

A fading transition surfaces as an alert: the hidden state likely changed, which is itself actionable information.

Injection: Before each prediction, the Oracle runs a local FAISS query (milliseconds) for the top-k relevant hypotheses, grouped by status. The Oracle states which hypotheses it relied on; these are tagged in the prediction record for future attribution.

7.3 Store State and Validation

The canonical store rebuild of 2026-07-07 (from the Oracle-branch snapshot) holds **56,347 hypotheses** with a monotone win-rate gradient across lifecycle tiers:

Tier	Win rate (canonical 2026-07-07)
strong	75.0%
emerging	68.2%
fading (n=107)	59.0%

The gradient (**strong 75.0% > emerging 68.2% > fading 59.0%**) validates the graduation and demotion criteria: they are correctly identifying higher- and lower-quality hypotheses. The store has grown roughly an order of magnitude since the system's first year while holding this gradient. An earlier repo snapshot (June 2026; 44,510 hypotheses) showed a steeper gradient (77.0% > 74.3% > 31.6%, with only 17 fading entries); the fading tier's win rate is sensitive to its small, changing membership, and we report the canonical rebuild as authoritative. Fading hypotheses are also hard-removed during dedup/rebuild once invalidated, so the tier's composition turns over by design.

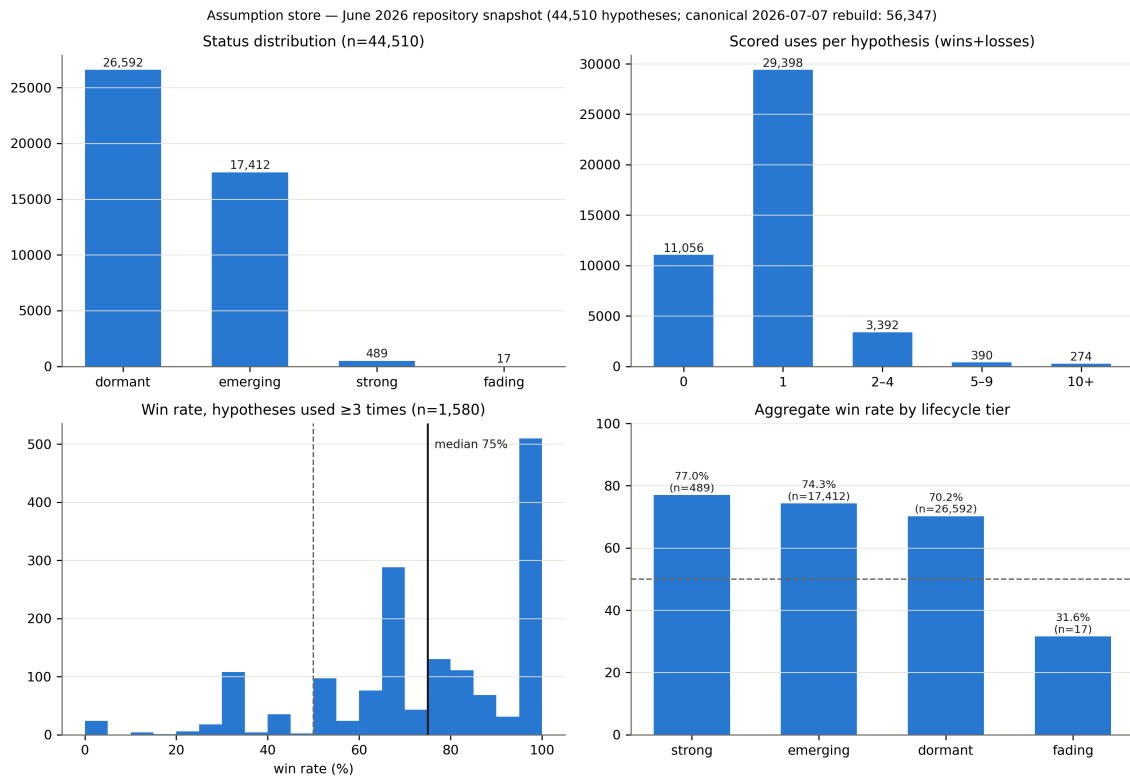


Figure 9: **Figure 9.** Assumption store, June 2026 repository snapshot (44,510 hypotheses; the canonical 2026-07-07 rebuild in the text holds 56,347), regenerated from the store metrics file by `analysis/generate_figures.py`. Clockwise from top-left: status distribution; scored uses per hypothesis (a long zero/single-use tail under a heavy head of high-use hypotheses); aggregate win rate by lifecycle tier (with tier sizes); and win-rate distribution for hypotheses used ≥ 3 times (median 75%, well right of the coin-flip line).

Example strong hypotheses (June 2026 snapshot, from the live store):

Text (truncated)	Domain	Uses	Win Rate
<i>The Hormuz closure is currently partial rather than total, allowing some tanker passage that would prevent the worst-case oil price spike</i>	cross-do-main	4,623	75%
<i>Meta's leadership has internally decided to release Llama 4 in March 2026 to compete with OpenAI's GPT-5 launch</i>	AI / tech	2,687	76%
<i>SK Hynix and Micron have confirmed HBM4 supply commitments enabling NVIDIA to publicly commit to Rubin specs</i>	semiconduc-tors	195	71%
<i>Skyrizi Q1 2026 prescription growth has continued without significant payer-driven access restrictions emerging</i>	pharma	185	85%

The fading tier represents active regime-change signals - world states that were reliable predictive context but have recently stopped working:

Text (truncated)	Domain	Uses	Recent Conf.
<i>FDA reviewers have not identified any new safety signal in the IV nimodipine formulation absent from the oral safety database</i>	FDA pharma	/ 117	0.22
<i>The INHALE-1 HbA1c non-inferiority margin was missed by a clinically meaningful amount, not borderline</i>	FDA	25	0.10

Each is a hidden-state assumption that scored well, then began losing as the underlying reality shifted - surfaced as an alert rather than silently deranked.

8. What Each Loop Corrects

Error Source	Loop	Mechanism	Evidence
No forecast points at an emerging regime	0	Abduction from world anomalies -> theses -> commissioned predictions	3 theses, 10 consequences commissioned (results resolve Sept 2026)
A validated insight doesn't circulate to the fleet	0 (outer)	Supported thesis -> assumption store; fading thesis -> Cornelius	promote/retract + Cornelius notify, built
Process inconsistency (occasion noise)	1	Structured reasoning templates, drift alerts	oracle-6: -14.7pp surprise
Conditional bias (domain-horizon variation)	1	Per-category adjustments, not global nudges	oracle-4: -12.8pp surprise
Systematic under/over-confidence	1	Bucket/confidence adjustments (when bias > 20% of error)	oracle-8: +7.7pp confidence, -11.9pp surprise
Feedback not absorbed	1	(open) advisory text delivered, per-agent uptake varies	oracle-2, oracle-7 moved against feedback
Wrong mental models	2	Failure -> Cornelius insight extraction	1,967 failure lessons delivered to the KB
Repeated cross-agent mistakes	3	Knowledge retrieval before every prediction	Fleet-wide Cornelius query before generating
Ungrounded hidden assumptions	4	Hypothesis extraction, scoring, injection	56,347 scored hypotheses (canonical 2026-07-07)
Regime-change blindness	4	Fading hypothesis alerts	107 active fading signals

9. Related Work

LLMs as forecasters. The human benchmark comes from the superforecasting tradition: structured process - reference classes, base rates, granular updating, and relentless Brier-score tracking - outperforms raw expertise (Tetlock and Gardner 2015). LLM-based systems have approached this bar: a retrieval-reason-aggregate pipeline nears the human crowd on real questions (Halawi et al. 2024), and an ensemble of twelve models is statistically indistinguishable from a 925-forecaster crowd (Schoenegger, Tuminauskaite, et al. 2024). But the gap has not closed: on a contamination-free standing benchmark, superforecasters (Brier 0.096) still significantly beat the best LLM (0.122) (Karger et al. 2024); unscaffolded GPT-4 performed no better than chance in a live tournament (Schoenegger and Park 2023); and results on judgmental forecasting tasks remain mixed (Abolghasemi et al. 2024). LLM assistance does causally improve *human* forecasters (+24-28% accuracy in a preregistered RCT) (Schoenegger, Park, et al. 2024), and consistency-based evaluation offers ground-truth-free quality signals - though optimizing consistency directly Goodharts without improving accuracy (Paleka et al. 2024). All of these treat the forecaster as a fixed artifact evaluated at a point in time; our contribution is the feedback architecture around the forecaster. Section 2.4 locates our fleet’s accuracy against these published reference points and against naive baselines computed on its own question set.

Calibration. Modern neural networks are systematically miscalibrated (Guo et al. 2017), and LLMs express overconfident verbalized probabilities (Xiong et al. 2023; Geng et al. 2024), a pathology traceable to preference tuning: reward models prefer confidently-stated answers regardless of quality (Leng et al. 2024), and calibration degrades during instruction tuning (Zhu et al. 2023). Remedies divide into elicitation-time methods - asking for probabilities the right way (Lin et al. 2022; Tian et al. 2023) or sampling and aggregating (Xiong et al. 2023) - and training-time methods that add calibration to the reward signal (Bani-Harouni et al. 2025; Damani et al. 2025), including RL trained directly on resolved forecasting outcomes (Turtel et al. 2025). Our approach is complementary to both: in-context feedback from resolved outcomes, without retraining - and, per Section 4, aimed primarily at noise, which the calibration toolkit does not target.

Noise, not bias. The judgment literature distinguishes systematic bias from noise - random scatter across interchangeable judges and occasions - and finds noise is often the larger error component in professional judgment (Kahneman et al. 2021, 2016), echoing the formal bias-variance decomposition of judgment error (Gigerenzer and Brighton 2009) and the classical finding that identical overall accuracy can decompose into very different error structures (Yates et al. 1989). Our fleet-wide measurement (93.5-99.7% noise) transplants this decomposition to LLM forecasting, where independent benchmarking has now replicated the phenomenon (Future-Search et al. 2025). Notably, the one randomized evaluation of a structured analytic technique (Analysis of Competing Hypotheses) found it did *not* improve accuracy and *increased* judgment inconsistency, while consistency itself strongly predicted accuracy (Dhami et al. 2019) - a caution that process interventions must be outcome-scored, not assumed to work, which is exactly what Loop 1’s before/after measurement provides.

Self-improvement without retraining. Iterative self-feedback (Madaan et al. 2023) and verbally-encoded episodic memory (Shinn et al. 2023) established that frozen models can improve

through language-level feedback; subsequent work distills experience into reusable insight pools (Zhao et al. 2024), persistent causal memories that transfer across tasks (Majumder et al. 2023), offline instruction learning from failures (Gupta et al. 2024), and cross-agent experience sharing (Qian et al. 2024) - capabilities organized theoretically by the CoALA framework, in which learning is writing to long-term memory (Sumers et al. 2023). The closest deployed analogue is verbal reinforcement inside a financial multi-agent system (Yu et al. 2024). Two failure modes motivate our design choices: self-refinement loops spontaneously reward-hack when the model judges its own output (J. Pan et al. 2024), and confident models resist self-correction through reflection alone (Tian et al. 2024). Every feedback signal in our system is therefore grounded in *resolved external outcomes*, never in the model’s own assessment - and, unlike the benchmark-episode settings above, the loops run continuously in deployment over months.

Retrieval and knowledge structures. Retrieval-augmented generation grounds generation in non-parametric memory that can be updated without retraining (Lewis et al. 2020; Gao et al. 2023); interleaving retrieval with reasoning steps (Trivedi et al. 2023) and retrieving actively at points of uncertainty (Jiang et al. 2023) both improve factuality. Knowledge graphs offer LLMs structured, updatable world knowledge (S. Pan et al. 2024). Cornelius differs in *what* it retrieves: distilled lessons from the system’s own prediction failures - a self-generating corpus - rather than an external document collection. Belief tracking in dialogue systems maintains scored distributions over hidden, non-stationary user goals (Henderson et al. 2014; Williams et al. 2016; Mrkšić et al. 2017); the assumption store is a forecasting analogue over hidden world states, scored by predictive track record.

Hypothesis generation and multi-agent evaluation. Intelligence-analysis tradecraft formalized the discipline of evaluating competing hypotheses against evidence (Heuer 1999). LLMs can propose novel, later-validated hypotheses (Qi et al. 2023), improve event prediction through abductive explanation of related events (Shi et al. 2023), and drive open-ended discovery when the search reward is Bayesian surprise (Agarwal et al. 2025); surveys of the emerging hypothesis-generation field stress grounding in structured knowledge and external validation (Alkan et al. 2025). Loop 0 differs from all of these in that every generated thesis is forced through the same falsifiable, honestly-resolved prediction machinery as an ordinary forecast. On the evaluation side, multi-agent debate improves LLM-based judging - but only with genuinely diverse perspectives (Chan et al. 2023; Tian et al. 2024) - and agent surveys identify environmental feedback as the key lever separating agentic from single-pass systems (Wang et al. 2024); Cleon instantiates this as a persistent, specialized meta-analyst rather than an ad-hoc judging panel.

10. Discussion and Limitations

The main finding is architectural: prediction error decomposes into distinct sources, and a deployed system improves when each source is routed to a matched correction mechanism - process guidance for noise, per-category adjustment for conditional bias, memory for wrong mental models, empirically-scored context for ungrounded assumptions, and abduction for the questions nobody asked.

The noise result reframes the calibration agenda. With 93.5-99.7% of error in the noise component, bucket recalibration - the default intervention - addresses at most a few percent of the problem. The practical implication for LLM forecasting systems is to measure the decomposition *first* and gate the intervention on it. That the same phenomenon appears in independent benchmarks (FutureSearch et al. 2025) suggests this is a property of current LLM forecasting generally, not of our fleet.

Feedback absorption is the open problem. The split fleet (Section 4.3) shows advisory-text feedback works often but not reliably: two of seven agents moved against direction-explicit advice. This echoes the human finding that structured process prescriptions do not automatically improve judgment (Dhami et al. 2019). Candidate explanations we cannot yet distinguish: (a) domain nonstationarity swamping the advisory signal (both regressing agents sit in fast-moving domains - markets and AI semiconductors); (b) prompt-level interference between calibration notes and other injected context; (c) genuine model-level resistance to confidence-raising instructions, the mirror image of the overconfidence documented at the population level (Xiong et al. 2023; Leng et al. 2024). Distinguishing these requires per-agent intervention experiments, which the current deployment does not randomize.

Limitations. (1) *No causal identification.* The Loop 1 evidence is a before/after comparison split at each agent’s first feedback delivery (2026-03-24; oracle-8, launched 2026-04-03, at 2026-04-05), not a randomized experiment; concurrent world drift is uncontrolled, and the internal reports’ uniform 2026-04-05 split yields slightly different - though qualitatively identical - deltas. (2) *Feedback absorption is unmodeled;* see above. (3) *Loop 0 is unscored:* zero of ten commissioned consequences resolve before 2026-09-19, so the generative loop’s value is architectural claim, not measured result, in this paper. (4) *Loop 2 delivery lags detection:* of 27,818 processed surprise entries, 1,967 insights have been delivered, and the retry queue (390) is rising - the measured effect of Loops 2-3 is therefore a lower bound on the design’s potential. (5) *Single model family:* all seven agents run the same base model (claude-opus-4-6), so fleet diversity is domain- and context-driven, not model-driven; inference settings and per-prediction inference dates are only partially recorded, short of full TRIPOD-LLM-style reporting. (6) *Snapshot governance:* during preparation of this paper we found the hypothesis-store metrics in the repository lagged the canonical store by roughly a month (44,510 vs 56,347 hypotheses; fading win rate 31.6% vs 59.0%) - we report the canonical rebuild and flag the earlier figure snapshot explicitly, but the incident illustrates that self-measuring systems need single-source-of-truth snapshot discipline. (7) *Resolution quality:* outcomes resolve against tiered web sources; the source-trust tier reduces but does not eliminate resolution error. (8) *No external benchmark evaluation.* All accuracy figures are on the fleet’s self-generated question set, which is imbalanced (68.3% of outcomes resolve true); Brier scores are only comparable on a common question set, so the published reference points in Section 2.4 are context, not comparison, and the in-sample base-rate baseline - the honest internal yardstick - is itself oracle-advantaged. Two agents (oracle-7, oracle-8) do not currently beat that baseline. Evaluating the fleet on a standing external benchmark such as ForecastBench (Karger et al. 2024) is the direct next step for accuracy claims.

Implications. The architecture suggests a general pattern for continuously improving AI forecasting systems: rather than waiting for retraining, structure the operational environment to route each type of error signal - and the absence of any signal at all - to an appropriate mechanism in real time. The September 2026 horizon resolutions will provide the first thesis-level calibration data for the generative loop; per-agent randomized feedback trials and delivery-pipeline hardening are the concrete next steps for the corrective loops.

11. Conclusion

The premise of this work is that judgment under uncertainty is measurable, and what is measurable is engineerable. A deployed fleet of seven forecasting agents keeps score on 73,552 resolved outcomes, decomposes its error into bias and noise, and routes each component to a matched correction mechanism: a meta-analyst (Cleon) that measures quality, coaches calibration, maintains an empirically-scored hidden-state store, and abduces falsifiable theses about what is worth forecasting; and a knowledge-base agent (Cornelius) that turns individual failures into fleet-wide lessons.

The loops interact: calibration identifies which domains have high occasion noise, prompting structured reasoning guidance (Loop 1); surprising failures are routed to Cornelius (Loop 2), where they become searchable knowledge retrieved before future predictions (Loop 3); the hidden-assumption store (Loop 4) provides empirically-grounded world-state context that calibration alone cannot supply; and synthesis (Loop 0) generates the theses the corrective loops would otherwise never be handed, feeding validated ones back into the same store.

The system's most distinctive epistemic object is the hidden-state layer: an explicit, continuously re-scored representation of the state of the world that is not publicly known but is assumed from evidence. Each of its 56,347 hypotheses is retained not because it is plausible but because of its measured predictive power - the monotone win-rate gradient (strong 75.0% > emerging 68.2% > fading 59.0%) shows the scoring separates reliable context from stale context, and a fading hypothesis is itself information: a regime-change alert. Together with the knowledge base's cross-domain lessons, this amounts to an image of the world maintained at the granularity of falsifiable assumptions - consulted before every new prediction.

Two boundaries keep the claim honest. First, such a system will never deliver black-and-white answers; irreducible uncertainty is its operating environment, and the achievable goal is calibration and demonstrable skill over naive baselines (fleet Brier skill score +0.105 over an in-sample base-rate forecaster; two agents still below that bar). Second, none of the measured improvement came from model scale: the base model was held fixed throughout deployment, so every delta reported in this paper is attributable to the orchestration and feedback layer around the model. We do not claim that layer outweighs raw model capability - that comparison was not run - but it is a lever available continuously, cheaply, and without retraining, and unlike a static model it compounds: insights accumulate, hypotheses re-score, theses resolve. The generative loop's first resolutions arrive in September 2026 and will extend the scorecard from forecasts to theses.

System and Data Availability

The deployed system’s live record is public: Brier HQ (<https://brierhq.ai>) publishes the fleet’s predictions, confidence scores, resolutions, Brier scores, and hidden-state hypotheses, and is updated continuously as predictions are generated and resolved - readers can audit the system’s ongoing performance beyond the snapshot reported here. Two of the system’s building blocks are open source: Trinity (<https://github.com/Abilityai/trinity>, Apache 2.0), the self-hosted runtime on which the agents are deployed, and Cornelius (<https://github.com/Abilityai/cornelius>), the knowledge-base agent template underlying the Loop 2/3 memory component. The fleet’s per-agent prompts, calibration notes, and the raw prediction CSV corpus analyzed in this paper remain private; all reported numbers are pinned to the snapshots documented in Section 2.1, and the analysis scripts that produced every figure and table accompany the manuscript.

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AI-use disclosure: see the accompanying `ai-disclosure.md` (drafting and citation management were AI-assisted; all system design, deployment, and analysis pipelines are the authors’; no AI system is an author).

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